

Substation Monitoring And Controlling Based on Microcontroller by using GSM Modem

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ABSTRACT:

The aim of this project is to acquire remote electrical parameters such as voltage, current and frequency and to submit these real-time values over the GSM network using GSM modem / phone along with the power station temperature. This project is also planned by running an Electromagnetic Relay to secure the electrical circuits. Whenever the electrical parameters exceed the predefined values, this Relay is triggered. Users can send SMS messaging commands to read the electrical parameters remotely. This was done on time setting) in SMS format. This onboard computer is able to interact effectively with the various sensors used. That internal memory is given to the controller to hold the code. Use this memory to dump some set of assembly instructions into the controller. And the controller's functionality depends on certain assembly instructions. The controller is designed using the language Embedded C.

1. INTRODUCTION

Electricity is a useful and extremely convenient form of energy. It plays an ever-increasing role in our industrialized capitalist society. The power systems are strongly non-linear, extremely large and dynamic networks. Such electric power systems are unified for cost-effective benefits, increased reliability and operational benefits. They are one of the most significant elements of both national and global infrastructure, and it contributes to substantial direct and indirect impacts on the economy and national security when these structures collapse. Components such as generators, pipes, transformers, loads, switches and compensators are used in a power network. However, the general design of modern power systems is a widely

distributed energy sources and charges. Electrical power systems can be divided into two sub-systems, i.e. transmission and distribution.

A transmission system's key function is to move electricity from electric generators to the customer location, while a distribution network provides the ultimate link between high voltage transmission systems and consumer services. The voltage level is then decreased using transformers, and power is transmitted to consumers via distribution systems for electric power. Power starts from the transmission grid at distribution substations where the voltage is stepped down (typically to less than 10kV) and supplied to business, residential, and industrial users by smaller distribution lines. A large number of distributed,

autonomously operated, capital-intensive assets consist of new electrical power systems like power transmission and distribution grids. Such assets include: 1.) Power plants, 2.) Transmission lines, 3.) Transformers and 4.) equipment for protection. Substations for electric utilities are used in both the transmission and distribution system and work separately to produce the electricity. Hundreds of miles of the power generated at the main stations is transported using transmission lines until they enter the substations. With the support of micro processors and controllers for continuous monitoring of sample concentrations, the actions of analysts at different time intervals, monitoring of voltage, current and temperature variations in distribution transformers at substations, number of work have been carried out over the past few years. Due to the increase in temperature at the distribution transformers, the current and voltage levels at the substations can vary drastically. The standard of power supplied to the customer can be inadequate because of this.

Therefore monitoring the current, voltage and additionally necessary parameters on the distribution side will help to improve both the output produced at the main station as well as the quality of power supplied on the customer side. It can also identify the break downs caused by heat, high temperature and overvoltage. If the temperature increase increases above the target temperature, then the control device can protect the distribution transformer by shutting down the machine. As discussed earlier, one of the biggest issues in the Electricity Board (EB) is the repair of a transformer. The

transformer is burned out during odd events due to the overload and short circuit in their winding, for certain reasons. The temperature of the oil is also increased due to the change in the amount of current flowing through its internal windings. This results in the distribution transformer experiencing a sudden change in voltage, current or temperature. Hence we are suggesting the automation of the EB substation delivery transformer. We consider voltage, current and temperature in automation as the parameters to be tracked as the transformer displays its peak sensitivity for the same. Hence, we build a microcontroller-based automation system that controls the transformer continuously. The transformer present in the substation which is turned off at the main station due to the microcontroller operation.

II.BLOCK DIAGRAM

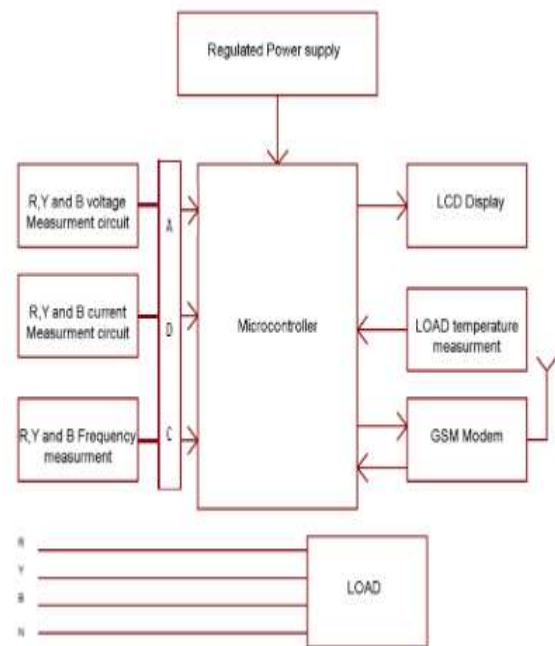


Fig 1: Block Diagram of the project

3.1 HARDWARE IMPLEMENTATION

ACS712 current sensor: 185mv/A output

sensitivity, supply voltage=8V, output voltage=8V,

storage temp=-65 °C to 165 °C

□ Pin no: VCC=+5V, OUT, GND

□ Apparatus: Load, relay (10A capacity), GSM, LCD

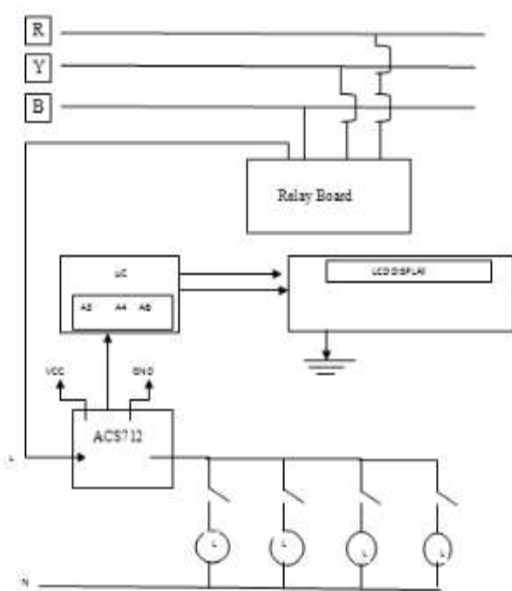


Fig 2: Block diagram representation of current measurement circuit

The circuit makes use of current sensor ACS712. Three current sensors are employed for the purpose of measuring currents in all 3 phases. One connection is done with the incoming phase wire and on the other connection is done with the wire on which the load is connected. For current the upper limit is set at 1.6A. The limit can be easily altered in case the load is changed. Here 4 lamps are connected as load. Limits are set such that the circuit operates

satisfactorily if 3 lamps are turned ON. When the 4th lamp is switched ON the relay operates and isolates the circuit. A message is sent through GSM stating the condition as overload. When the 4th lamp is switched OFF, the relay again operates to connect the load with the supply. Again a message is sent stating that the power has been restored.

3.2 Frequency Measurement circuit:

□ Apparatus: IC 555 Timer, 220K POT, +5V DC, GSM, LCD display,

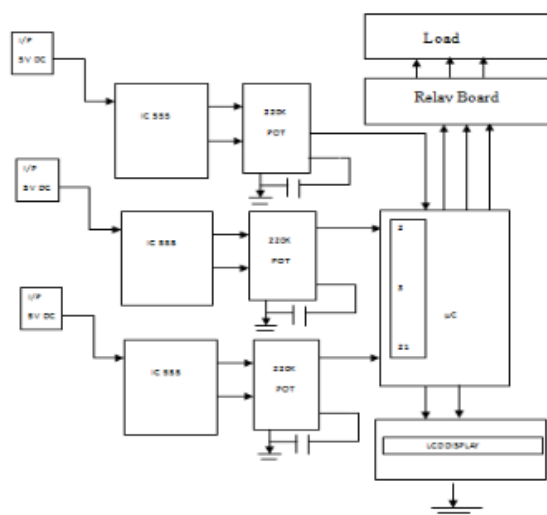


Fig -3: Block diagram representation of Frequency measurement circuit

The circuit makes use of IC555 timer. The IC555 is made to work in Astable mode. Three IC's are employed for the purpose of varying frequency in all 3 phases (As it is not possible to change the supply frequency). The lower limit for frequency is 48Hz and the upper limit is kept as 52Hz for the purpose of the project.. The limit can be easily altered as per the requirement. The circuit operates satisfactorily if the frequency is within the limits. The relay operates and

isolates the circuit if the frequency falls outside the above limits. A message is sent through GSM stating the sensed condition as under frequency or over frequency. As the frequency comes back within the limit, the relay connects the load with the supply.

3.3 Temperature Measurement circuit:

□ Apparatus: LM 35 Temperature Sensors, Load, LCD Display, Arduino

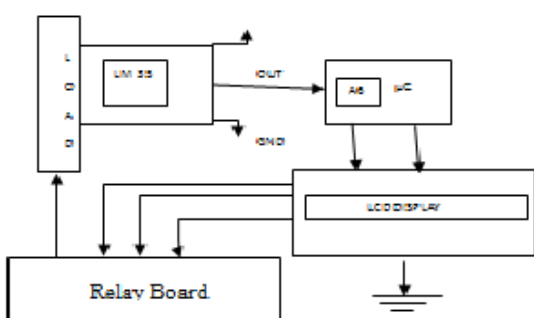


Fig -4: Block diagram representation of Temperature measurement circuit

The circuit makes use of temperature sensor LM35. Only one sensor is employed for the purpose of measuring the temperature. It has 3 pins namely VCC, Output & ground which are connected with the controller. The limit is kept at 80°C. The limit can be easily altered as per the requirement. The relay operates and isolates the circuit if the temperature exceeds 80°C. A message is sent through GSM stating the condition as overheating of the equipment with which the sensor is connected. When the temperature comes within the limits, the relays connect to the equipment with the supply. A message is sent stating that the power has been restored.

IV.RESULTS



Fig 5: Actual Frequency measurement circuit



Fig -6: Actual Temperature measurement circuit

V.CONCLUSION & FUTURE SCOPE

Our project microcontroller based substation monitoring and control system with iot module. But in future we can also add gsm modem is mainly intended to operate the devices like fans, lights, motors etc., through a gsm based mobile phone. The system has a gsm modem, temperature, current, voltage sensors and the devices to be operated through the switches like relay which are interfaced to the micro controller. The micro controller is programmed in such a way that if a particular fixed format of sms

is sent to gsm modem from mobile phone, which is fed as input to the micro controller which operates the appropriate devices. A return feedback message will be sent to the mobile from gsm modem. The temperature at the place where devices are being operated can be known. In future we can use this project in several applications by adding additional components to this project. This project can be extended by using gprs technology, which helps in sending the monitored and controlled data to any place in the world. The temperature controlling systems like coolant can also use in places where temperature level should be maintained. By connecting wireless camera in industries, factories etc we can see the entire equipments from our personal computer only by using gprs and gps technology. The monitoring and controlling of the devices can be done from the personal computer and we can use to handle so many situations. By connecting temperature sensor, we can get the temperature of dangerous zones in industries and we can use personal computer itself instead of sending human to there and facing problems at the field. The temperature sensor will detect the temperature and it gives information to the micro controller and micro controller gives the information to the mobile phone from that we can get the data at pc side.

VI. REFERENCES

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